Brain Connectivity Analysis Methods Overview

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User prompt: references Connectivity Analysis Methods

Response: Key scientific literature and foundational reviews for **brain connectivity analysis methods** are organized below by methodological focus (Functional, Effective, Structural, Graph Theoretical/Connectomics, and Multi-modal integration).

Overview of Connectivity Categories

Brain connectivity is broadly categorized into three core modalities (Friston, 2011; Rossini et al., 2019):

1. **Structural (Anatomical) Connectivity:** Physical or axonal pathways interconnecting neural elements, typically measured in vivo using diffusion-weighted MRI (dMRI) and diffusion tractography.
2. **Functional Connectivity (FC):** Statistical dependencies and temporal correlations between neurophysiological events in spatially distant brain regions (e.g., seed-based correlation, independent component analysis, or phase synchrony) without explicitly assuming causal directionality.
3. **Effective Connectivity (EC):** The direct or causal influence that one neural system exerts over another, parameterized using explicit mechanistic or biophysical models (e.g., Dynamic Causal Modeling or Granger Causality).

Core Methodological Paradigms

Graph Theory & Network Topology

Representing brain systems as complex network graphs—where nodes represent parcellated brain regions and edges represent structural or functional links—enables quantitative characterization of topology (Bullmore & Sporns, 2009). Commonly evaluated graph metrics include path length, clustering coefficient, small-worldness, modularity, and node centrality (Rubinov & Sporns, 2010).

Computational & Signal Processing Techniques

To decompose high-dimensional neuroimaging time series (from fMRI, EEG, or MEG), blind source separation algorithms such as Independent Component Analysis (ICA) and multivariate decomposition methods are frequently employed to isolate intrinsic network signals from artifactual noise (Lang et al., 2012).

Multi-modal & Clinical Applications

Simultaneous recording modalities (such as EEG-fMRI) combine high temporal and spatial resolutions to map dynamic network reorganizations and identify network biomarkers across

neurological and psychiatric conditions (Rossini et al., 2019; Yen et al., 2023).

Key Methodological References

Category	Primary Focus	Key Reference
Theoretical Framework	Functional vs. Effective Connectivity Foundations	Friston (2011)
Network Architecture	Graph Theoretical Analysis & Brain Topology	Bullmore & Sporns (2009)
Network Quantification	Network Measures & Brain Connectivity Toolbox	Rubinov & Sporns (2010)
Consensus Review	Multi-modal Connectivity Analysis (IFCN Consensus)	Rossini et al. (2019)
Signal Processing	Decomposition & Survey of FC/EC Algorithms	Lang et al. (2012)
Neuroimaging Advances	Modern Functional Neuroimaging & Clinical Biomarkers	Yen et al. (2023)

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